Cardiometabolic Digital Twin

Predictive Glycemic Excursion & Counterfactual Intervention System for Type 2 Diabetes

1 LAYER

Data Generation

Synthetic Synthea-style EHR cohort — 300 T2DM patients
Bergman Minimal Model ODE + Dalla Man gut absorption kinetics
300 x 14-day wearable streams @ 5-min (CGM, HR, HRV, steps, sleep)

2 LAYER

Feature Pipeline

4-hour sliding window (48 readings), stepped every 30 min
47 features = 18 static EHR + 29 dynamic wearable signals
Patient-level split — 198,000 samples, zero data leakage

3 LAYER

Machine Learning & Explainability

Fused XGBoost regressor — 2-hour glucose forecast (R2 = 0.805)
Class-weighted adverse-event classifier (AUROC = 0.976)
SHAP TreeExplainer — per-prediction clinical attributions

4 LAYER

Digital Twin Engine

Stateful per-patient twin initialized from the EHR profile
predict() — 2h forecast, risk level, top-5 SHAP factors
simulate() — deterministic Bergman ODE What-If counterfactuals

5 LAYER

FastAPI REST Backend

Async API with per-patient twin caching, served on port 8000
/patients /patient/{id} /state /timeline /predict /simulate
Serves the dashboard and OpenAPI (Swagger) documentation

6 LAYER

Clinician Dashboard

Patient 360 profile - SVG 2-hour adverse-risk gauge
Dual-axis CGM / heart-rate timeline with 2h forecast overlay
SHAP attribution bars - What-If intervention simulator (live)